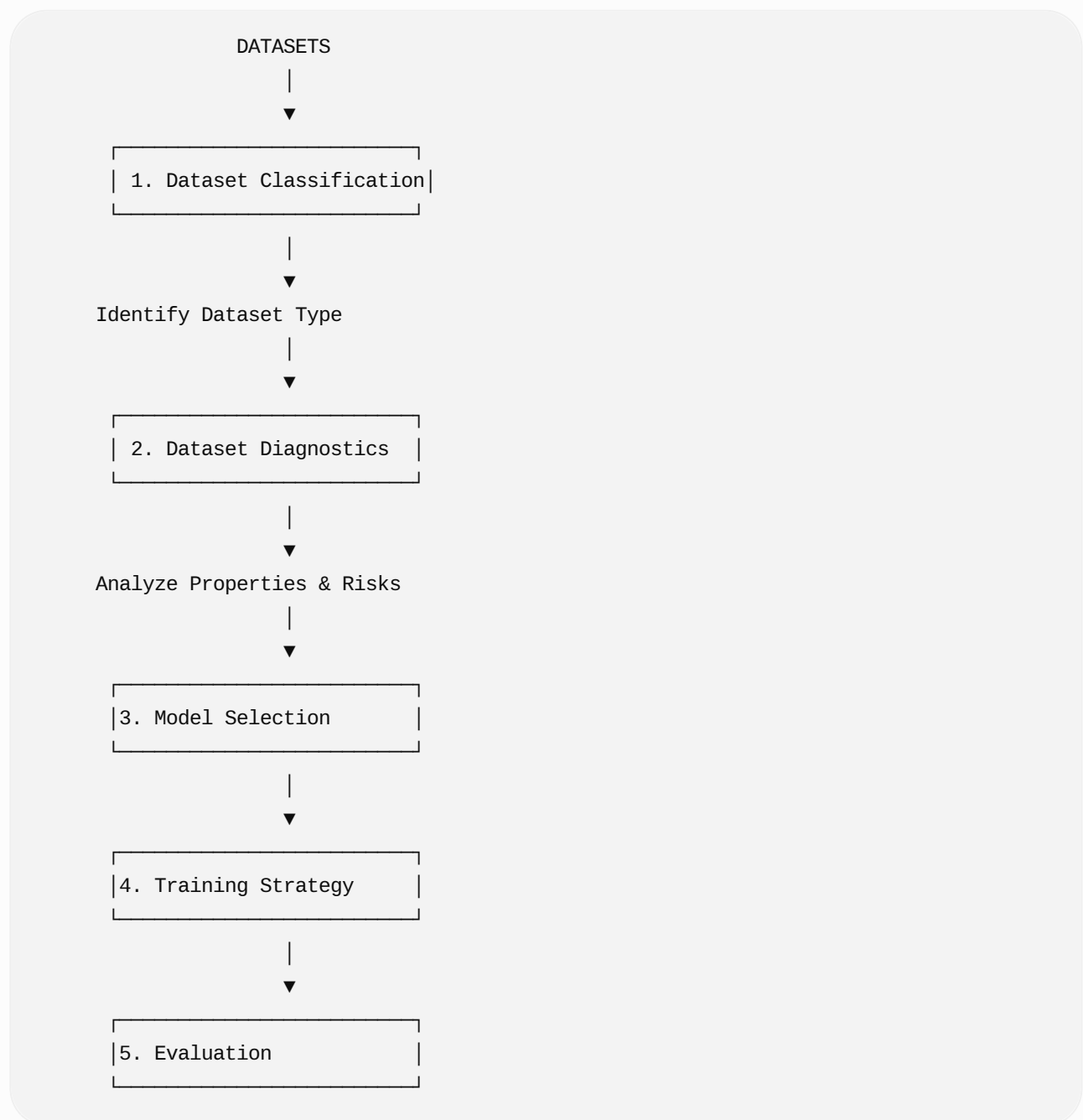

Complete Learning Roadmap



The problem with the previous table

That table is only valid for **Image datasets**.

If tomorrow someone uploads

- CSV
- Excel
- JSON
- Audio
- Video
- Medical Records
- SQL Database
- Time-series
- Text Corpus

then almost **70% of that image-specific audit no longer applies**.

For example

Image Dataset

- ✓ Resolution
 - ✓ RGB Channels
 - ✓ Blur
 - ✓ Noise
 - ✓ Aspect Ratio
-

Text Dataset

- ✗ Resolution
- ✗ Blur
- ✗ RGB

Instead check

- Vocabulary Size
- Average Sentence Length
- OOV Rate

- Language Distribution
 - Stopwords
 - Token Distribution
-

Audio Dataset

Instead check

- Sample Rate
 - Bit Depth
 - Duration
 - Silence Ratio
 - Background Noise
 - SNR
 - Frequency Spectrum
-

Time-Series Dataset

Instead check

- Sampling Interval
 - Missing Time Steps
 - Trend
 - Seasonality
 - Stationarity
 - Autocorrelation
 - Drift
-

Tabular Dataset

Instead check

- Missing Values
- Correlation
- Cardinality
- Skewness
- Outliers
- Data Types
- Duplicate Rows

- Target Leakage
-

So what should come next?

I would build the system like this.

Stage 1

Dataset Classification

(User uploads dataset)

↓

System identifies

- Structured
- Unstructured
- Semi-Structured

↓

Then identifies

- Image
 - Text
 - Audio
 - Video
 - Time-Series
 - Tabular
 - Multimodal
-

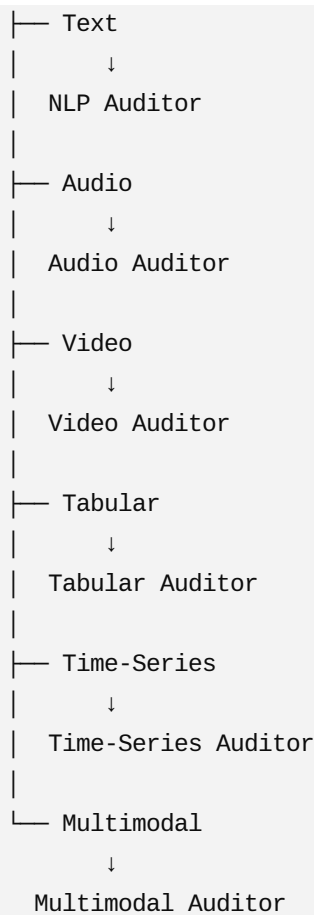
Stage 2

Dataset-Specific Auditor

Different audit engine for every dataset type.

Dataset





Stage 3

Every auditor has its own checklist.

Image Auditor

- Resolution
- Aspect Ratio
- Blur
- Noise
- Contrast
- Brightness
- Class Balance
- Duplicates
- Leakage

Text Auditor

- Vocabulary Size
- Average Length
- Language Detection

- Duplicate Documents
 - Label Balance
 - Token Distribution
 - OOV Words
 - Readability
-

Audio Auditor

- Sample Rate
 - Duration
 - Noise
 - Silence
 - Frequency Spectrum
 - Loudness
 - Clipping
-

Video Auditor

- FPS
 - Resolution
 - Motion Blur
 - Scene Changes
 - Duration
 - Frame Quality
 - Compression
-

Tabular Auditor

- Missing Values
 - Duplicate Rows
 - Outliers
 - Correlation
 - Multicollinearity
 - Feature Importance
 - Target Leakage
 - Class Imbalance
-

Time-Series Auditor

- Missing Timestamps
- Drift
- Trend
- Seasonality
- Stationarity
- Autocorrelation
- Change Points

Then comes the most powerful layer

Instead of just saying:

"Missing values = 20%"

The system should explain **what that means for model design**.

For example:

Finding

↓

20% Missing Values

↓

Risk

↓

Tree models may tolerate this,
Linear models may require imputation.

↓

Recommendation

↓

Median Imputation

↓

Affected Models

↓

XGBoost
CatBoost
Random Forest

↓

Validation

↓

Compare RMSE before and after imputation

Every audit finding should follow this chain:

Measure

↓

Interpret

↓

Risk

↓

Recommendation

↓

Affected Models

↓

Validation Metrics

This is the direction I would recommend if your goal is to build an AI Dataset Auditor.

Rather than creating one generic auditor, build a **Universal Dataset Auditor** that first **classifies the uploaded dataset**, then automatically launches the **appropriate specialized audit pipeline** (Image, Tabular, Text, Audio, Video, Time-Series, or Multimodal). This makes the system extensible, accurate, and much more valuable because every dataset type is evaluated using the diagnostics and recommendations that are actually relevant to it. That architecture is much closer to what would be expected in a research-grade or enterprise-grade auditing platform.